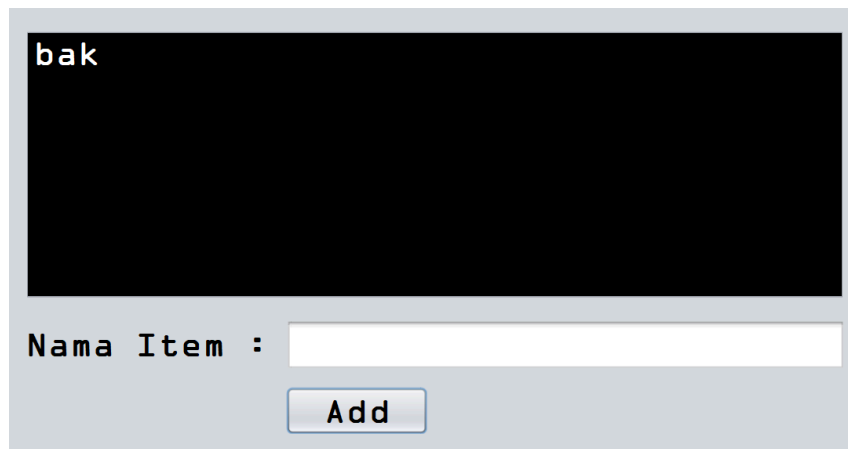


PROJECT 1

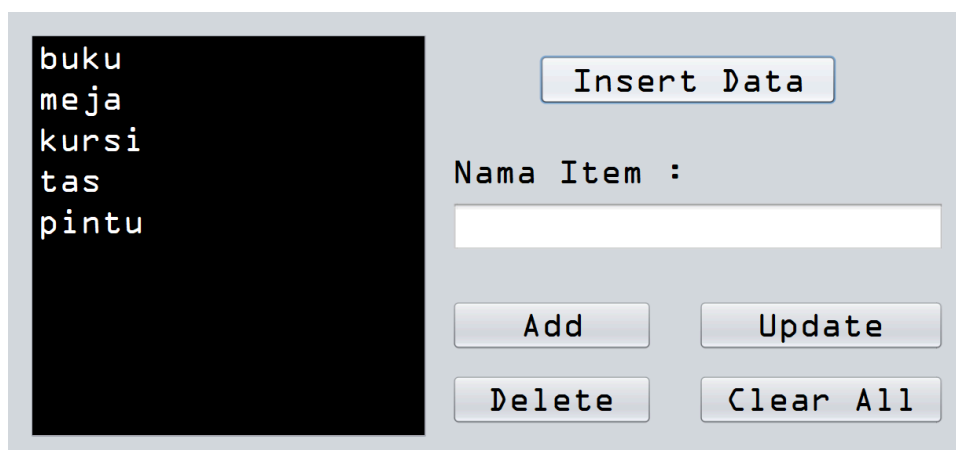
Add item bak



A screenshot of a simple application window with a light gray border. Inside, there is a large black rectangular area at the top. Below it, the text "Nama Item :" is followed by a white text input field. At the bottom center, there is a button labeled "Add".

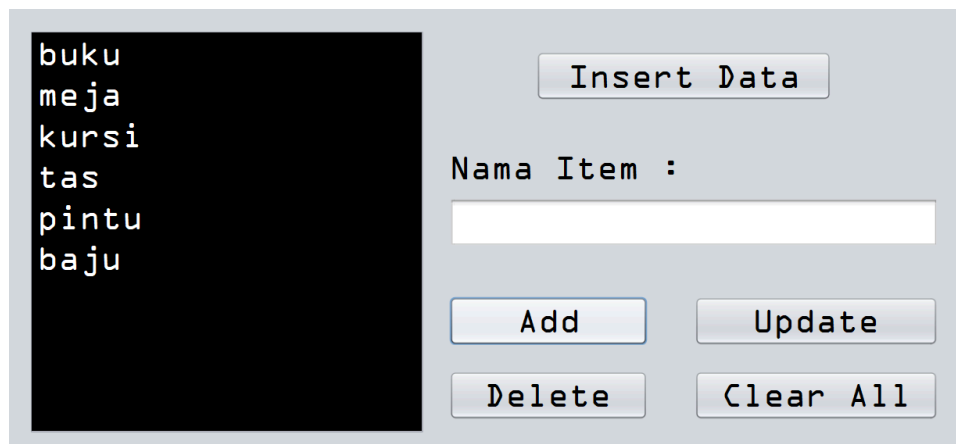
PROJECT 2

Insert Data



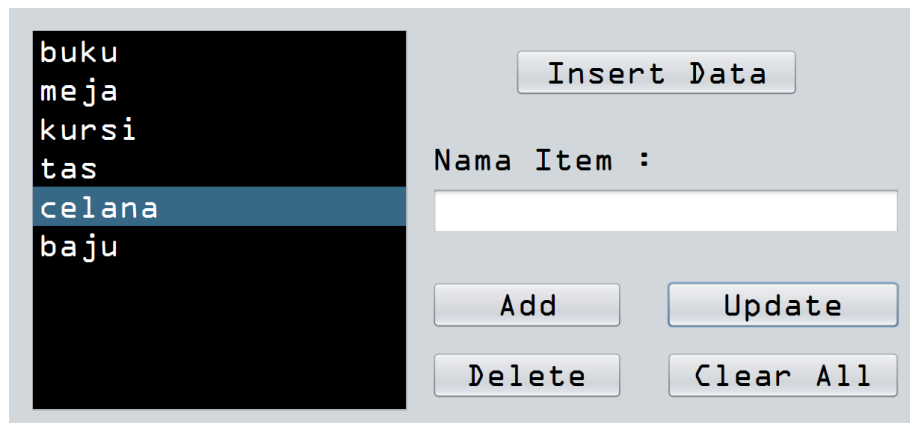
A screenshot of a more complex application window. On the left, there is a black rectangular area containing the text "buku", "meja", "kursi", "tas", and "pintu" stacked vertically. To the right of this area, there is a button labeled "Insert Data". Below the button, the text "Nama Item :" is followed by a white text input field. At the bottom, there are four buttons arranged in a 2x2 grid: "Add", "Update", "Delete", and "Clear All".

Add item baju



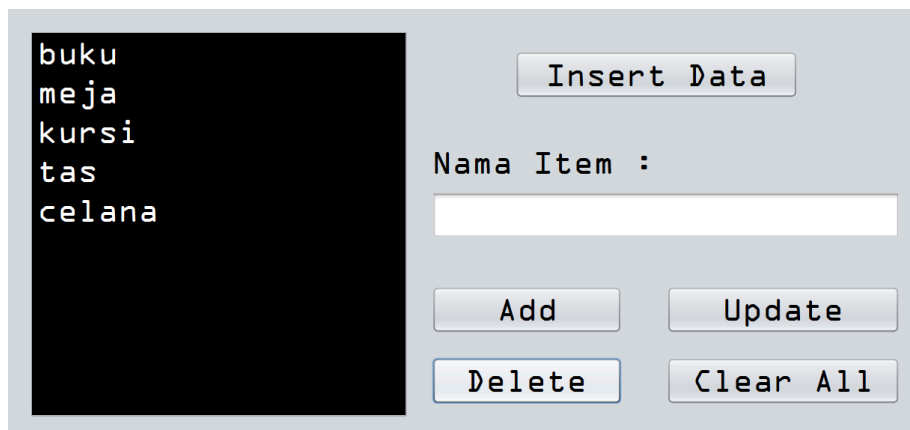
A screenshot of the same application window as in the previous image, but with an additional item, "baju", added to the black rectangular area on the left. The list now contains "buku", "meja", "kursi", "tas", "pintu", and "baju". The other elements, including the "Insert Data" button and the bottom buttons, remain the same.

Update item pintu jadi celana



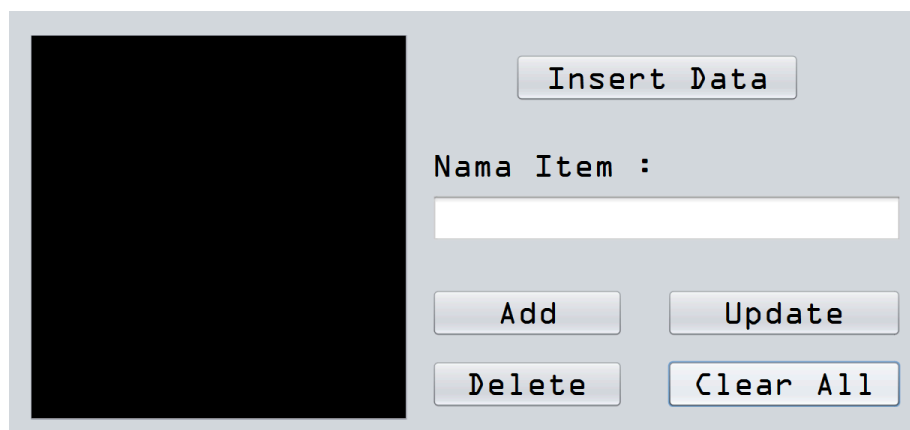
The screenshot shows a web application interface with a light gray background. On the left, there is a black rectangular area containing a list of items in white text: 'buku', 'meja', 'kursi', 'tas', 'celana', and 'baju'. The item 'celana' is highlighted with a blue background. To the right of this list is a control panel. At the top of the panel is a button labeled 'Insert Data'. Below it is a label 'Nama Item :' followed by a white text input field. At the bottom of the panel are four buttons arranged in a 2x2 grid: 'Add', 'Update', 'Delete', and 'Clear All'. The 'Update' button is highlighted with a blue border.

Delete item baju



This screenshot shows the same web application interface as the first one. In the black list area, the item 'baju' is now highlighted with a blue background, while 'celana' is no longer highlighted. In the control panel on the right, the 'Delete' button at the bottom left is now highlighted with a blue border.

Clear All



This screenshot shows the web application interface after the 'Clear All' action. The black rectangular area on the left is now completely empty. In the control panel on the right, the 'Clear All' button at the bottom right is highlighted with a blue border.

PROJECT 3

Insert Data

buku
meja
kursi
tas
pintu
buku
meja
kursi
tas
pintu

Insert Data Data tersimpan = 0

Save Data

Nama Item :

Add Update

Delete Clear All

Save Data

buku
meja
kursi
tas
pintu
buku
meja
kursi
tas
pintu

Insert Data Data tersimpan = 10

Save Data

Nama Item :

Add Update

Delete Clear All

Add item baju

buku
meja
kursi
tas
pintu
buku
meja
kursi
tas
pintu
baju

Insert Data Data tersimpan = 10

Save Data

Nama Item :

Add Update

Delete Clear All

Update item kursi menjadi celana

buku
meja
kursi
tas
pintu
buku
meja
celana
tas
pintu
baju

Insert Data Data tersimpan = 10

Save Data

Nama Item :

Add Update

Delete Clear All

Delete item baju

buku
meja
kursi
tas
pintu
buku
meja
celana
tas
pintu

Insert Data

 Data tersimpan = 10

Save Data

Nama Item :

Add

Update

Delete

Clear All

Clear All

Insert Data

 Data tersimpan = 10

Save Data

Nama Item :

Add

Update

Delete

Clear All

PROJECT 4

Save to List

buku
meja
kursi
tas
pintu

Save To List ->

 Data tersimpan = 5

<- Insert Data List

Save To Set ->

 Data tersimpan = 0

<- Insert Data Set

Save To Map ->

 Data tersimpan = 0

<- Insert Data Map

Nama Item :

Clear All

Add

Update

Delete

Insert Data List

buku	Save To List ->	Data tersimpan = 5
meja	<- Insert Data List	
kursi	Save To Set ->	Data tersimpan = 0
tas	<- Insert Data Set	
pintu	Save To Map ->	Data tersimpan = 0
buku	<- Insert Data Map	
meja		
kursi		
tas		
pintu		
Nama Item :		
Clear All	Add	Update Delete

Save to Set

buku	Save To List ->	Data tersimpan = 5
meja	<- Insert Data List	
kursi	Save To Set ->	Data tersimpan = 5
tas	<- Insert Data Set	
pintu	Save To Map ->	Data tersimpan = 0
buku	<- Insert Data Map	
meja		
kursi		
tas		
pintu		
Nama Item :		
Clear All	Add	Update Delete

Clear All, Insert Data Set

buku	Save To List ->	Data tersimpan = 5
kursi	<- Insert Data List	
tas	Save To Set ->	Data tersimpan = 5
meja	<- Insert Data Set	
pintu	Save To Map ->	Data tersimpan = 0
	<- Insert Data Map	
Nama Item :		
Clear All	Add	Update Delete

Save to Map

The screenshot shows a software interface with a black terminal window on the left containing the text: buku, kursi, tas, meja, pintu, buku, kursi, tas, meja, pintu. To the right of the terminal are three sets of buttons. The first set has 'Save To List ->' and '<- Insert Data List', with 'Data tersimpan = 5' to the right. The second set has 'Save To Set ->' and '<- Insert Data Set', also with 'Data tersimpan = 5' to the right. The third set has 'Save To Map ->' and '<- Insert Data Map', with 'Data tersimpan = 10' to the right. Below these is a label 'Nama Item :' followed by an empty text input field. At the bottom are four buttons: 'Clear All', 'Add', 'Update', and 'Delete'.

buku
kursi
tas
meja
pintu
buku
kursi
tas
meja
pintu

Save To List -> Data tersimpan = 5
<- Insert Data List

Save To Set -> Data tersimpan = 5
<- Insert Data Set

Save To Map -> Data tersimpan = 10
<- Insert Data Map

Nama Item :

Clear All Add Update Delete

Clear All, Insert Data Map

This screenshot is identical to the previous one, but the '<- Insert Data Map' button is highlighted with a blue border, indicating it is the active or selected action.

buku
kursi
tas
meja
pintu
buku
kursi
tas
meja
pintu

Save To List -> Data tersimpan = 5
<- Insert Data List

Save To Set -> Data tersimpan = 5
<- Insert Data Set

Save To Map -> Data tersimpan = 10
<- Insert Data Map

Nama Item :

Clear All Add Update Delete

Add item meja

This screenshot is identical to the previous ones, but the 'Add' button at the bottom is highlighted with a blue border, indicating it is the active or selected action.

buku
kursi
tas
meja
pintu
buku
kursi
tas
meja
pintu
meja

Save To List -> Data tersimpan = 5
<- Insert Data List

Save To Set -> Data tersimpan = 5
<- Insert Data Set

Save To Map -> Data tersimpan = 10
<- Insert Data Map

Nama Item :

Clear All Add Update Delete

Update item meja menjadi baju

buku
kursi
tas
meja
pintu
buku
kursi
tas
meja
pintu
baju

Save To List -> Data tersimpan = 5

<- Insert Data List

Save To Set -> Data tersimpan = 5

<- Insert Data Set

Save To Map -> Data tersimpan = 10

<- Insert Data Map

Nama Item :

Clear All Add Update Delete

Delete item buku (item pertama)

kursi
tas
meja
pintu
buku
kursi
tas
meja
pintu
baju

Save To List -> Data tersimpan = 5

<- Insert Data List

Save To Set -> Data tersimpan = 5

<- Insert Data Set

Save To Map -> Data tersimpan = 10

<- Insert Data Map

Nama Item :

Clear All Add Update Delete

JDBC_MHS_SERVICE_GUI

```
3 • use pboj;
4 • drop table mahasiswa;
5
6 • create table mahasiswa (
7   id int auto_increment primary key,
8   nama varchar(50));
9
10 • insert into mahasiswa (nama) values
11   ('sangyeon'),
12   ('jacob'),
13   ('younghoon');
14
15 • select * from mahasiswa;
```

Result Grid

Filter Rows:

Edit: Export/Import: Wrap Cell Content: 15

	id	nama
▶	1	sangyeon
	2	jacob
	3	younghoon
*	NULL	NULL

Ambil Semua Data

ID	Nama
1	sangyeon
2	jacob
3	younghoon

Kelola Data Mahasiswa

Ambil Semua Data

Nama :

Tambahkan

☐

Edit

☐

Hapus

Reset Indeks

Tambahkan sunwoo

ID	Nama
1	sangyeon
2	jacob
3	younghoon
5	sunwoo

Kelola Data Mahasiswa

Ambil Semua Data

Nama :

Tambahkan

☐

Edit

☐

Hapus

Reset Indeks

Edit sunwoo menjadi eric

ID	Nama
1	sangyeon
2	jacob
3	younghoon
5	eric

Kelola Data Mahasiswa

Ambil Semua Data

Nama :

Tambahkan

☐

Edit

☐

Hapus

Reset Indeks

Hapus eric

ID	Nama
1	sangyeon
2	jacob
3	younghoon

Kelola Data Mahasiswa

Ambil Semua Data

Nama :

Tambahkan

Edit

Hapus

Reset Indeks

Reset Index, Tambahkan sunwoo

ID	Nama
1	sangyeon
2	jacob
3	younghoon
4	sunwoo

Kelola Data Mahasiswa

Ambil Semua Data

Nama :

Tambahkan

Edit

Hapus

Reset Indeks

```
3 • use pboj;
4 • drop table mahasiswa;
5
6 • create table mahasiswa (
7   id int auto_increment primary key,
8   nama varchar(50));
9
10 • insert into mahasiswa (nama) values
11   ('sangyeon'),
12   ('jacob'),
13   ('younghoon');
14
15 • select * from mahasiswa;
```

Result Grid

	id	nama
▶	1	sangyeon
	2	jacob
	3	younghoon
	4	sunwoo
•	NULL	NULL